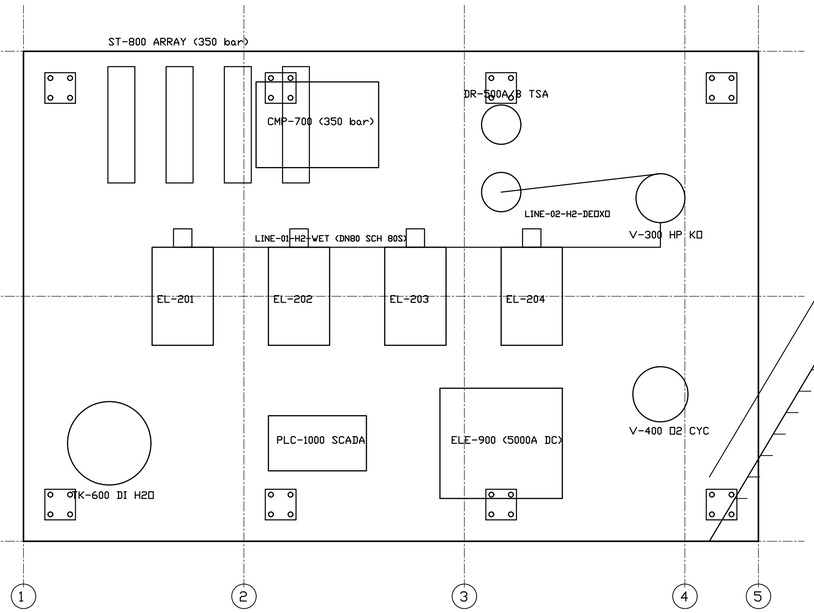
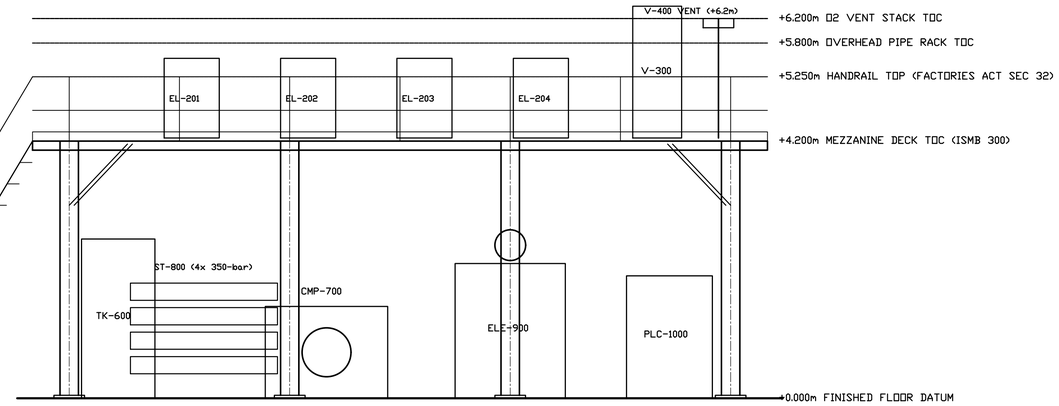
VIEW A-A: MASTER FACILITY PROCESS PLOT PLAN (SCALE 1:50)

12.0 m x 8.0 m MODULAR PROCESS SKID & ISMB 300 STRUCTURAL MEZZANINE



VIEW B-B: PLANT LONGITUDINAL ELEVATION & ISMB 300 STEEL FRAME (SCALE 1:50)

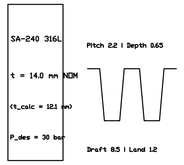
DATUM EL +0.000m FINISHED FLOOR TO ROOF VENT TOP EL +6.200m (IS 800:2007 & FACTORIES ACT 1948)



COMMERCIAL PLANT SUBSYSTEM MASTER SCHEDULE & STATUTORY SPECIFICATIONS (100 TPD)

TAG	UNIT DESCRIPTION	DESIGN CAPACITY / SPECS	STATUTORY CODE & MOC
STR-100	Modular Mezzanine Skid & Pipe Rack	12.0 x 8.0 x 5.8 m, Factories Act Handrails	IS 800:2007 / Factories Act 1948 / IS 875
EL-200	Commercial PEM Electrolyzer Bay	4x 50-Cell Stacks, 5000A DC Busbars	ISO 22734 / ASTM B265 Gr. 2 Ti / Nafion 115
V-300	HP Hydrogen Knockout Drum	Ø800 x 2650 mm, 30 bar, Demister Pad	ASME Sec VIII Div 1 (UG-27) / PESO SMPV 2016
V-400	Anode O2 Cyclone Degasser	Ø900 x 2800 mm, 6.2m Roof Vent Stack	SS316L / API 520 / NFPA 68 Deflagration
DR-500	Catalytic DeOxo & Dual 13X TSA Dryers	Twin 13X Towers, -70°C Dew Point	ISO 14687:2019 Grade D (<1 ppm O2, <1 ppm H2O)
WTR-600	DI Water Conditioning & Feed Pumps	3000L Storage, 2x 100% ANSI Pumps, PHE	ASTM D1193 Type 1 (>18.2 MΩ·cm) / ASME BPE
CMP-700	3-Stage HP Diaphragm Compressor	30 to 350 bar, 45 kW ATEX Zone 1 VFD	API 618 5th Ed / ASME B31.3 / Class 2500
ST-800	HP Hydrogen Buffer Storage Array	4x 350-bar Forged Accumulators, 75mm Wall	ASME Sec VIII Div 3 KD-250 / PESO Gas Cylinders
ELE-900	Substation Transformer & Rectifier	11 kV Step-Down, 5000A DC Power Center	IS 2026 / IEC 60076 / IEEE 519 (<3% THD)
PLC-1000	Siemens SCADA Automation Center	S7-1500F Fail-Safe, 21.5" Touch HMI	IEC 61511 / IEC 61508 SIL-2 HIPPS (<1.4s Trip)
PIP-1100	Interconnecting Process Piping	Schedule 80S / Class 2500 Piping Network	ASME B31.3 Process / ASME B31.12 Hydrogen

DETAIL C-C: UGBE7AIL D-D: TI CHANNELS



GENERAL ENGINEERING & STATUTORY COMPLIANCE NOTES:

1. CIVIL & STRUCTURAL: IS 800:2007 (STEEL STRUCTURES) AND IS 875 (PART 3 WIND LOADS).
2. OCCUPATIONAL SAFETY: ALL WALKWAYS, STAIRWAYS, HANDRAILS (1050mm) AND TOEBOARDS (150mm) STRICTLY CONFORM TO THE FACTORIES ACT, 1948 (SECTIONS 32 AND 33).
3. PRESSURE VESSELS: V-300 & V-400 MANUFACTURED UNDER ASME BPVC SECTION VIII DIV 1.
4. HIGH PRESSURE STORAGE: ST-800 ACCUMULATORS COMPLY WITH ASME SEC VIII DIV 3 (PART KD) & PETROLEUM AND EXPLOSIVES SAFETY ORGANIZATION (PESO) SMPV (UNFIRED) RULES 2016.
5. PROCESS PIPING: HYDROGEN LINES COMPLY WITH ASME B31.12; DI WATER/O2 COMPLY WITH ASME B31.3.
6. FUNCTIONAL SAFETY: SIL-2 HIPPS (<1.4s OPTICAL TRIP) GOVERNED BY IEC 61511 / IEC 61508.
7. HYDROGEN FUEL SPEC: DEOXO-TSA (DR-500) GUARANTEES ISO 14687:2019 GRADE D PURITY.
8. ALL DIMENSIONS IN MM UNLESS NOTED. WELDING PER ASME SECTION IX (GTAW/GMAW).

JADAVPUR UNIVERSITY & SISTER NIVEDITA UNIVERSITY

DEPT. OF CHEMICAL ENGINEERING | DEPT. OF MICROBIOLOGY & BIOTECH

COMMERCIAL GREEN HYDROGEN PRODUCTION COMPLEX (100 TPD)

11-SUBSYSTEM GENERAL ARRANGEMENT & ELEVATION SCHEMATIC

GENERAL ARRANGEMENT, ISMB 300 ELEVATION & PROCESS PLOT PLAN

DWG NO: DWG-PEM-H2-COMMERCIAL-PLANT-002 REV: 05 (COMMERCIAL)	SCALE: 1:50 / 1:25 PROJECTION: 3RD ANGLE	SHEET: 1 OF 2 SIZE: ISO A1 (841x594)
LEAD ENG: SOURISH SENAPATI (CHEM ENG, JU) BIOTECH: SRESTHA BISWASH (MICROBIO, SNU)	COMPLIANCE: FACTORIES ACT 1948 / ASME SEC VIII / PESO SMPV 2016	IS 800 STATUS: APPROVED FOR TENDER